

Photochemistry and electron-transfer mechanism of transition metal oxalato complexes excited in the charge transfer band

Jie Chen[†], Hua Zhang[†], Ivan V. Tomov[†], Xunliang Ding[†], and Peter M. Rentzepis^{†§}

[†]Department of Chemistry, University of California, Irvine, CA 92697; and [‡]Institute of Low Energy Nuclear Physics, Beijing Normal University, Beijing 100875, China

Contributed by Peter M. Rentzepis, August 7, 2008 (sent for review June 10, 2008)

The photoredox reaction of trisoxalato cobaltate (III) has been studied by means of ultrafast extended x-ray absorption fine structure and optical transient spectroscopy after excitation in the charge-transfer band with 267-nm femtosecond pulses. The Co–O transient bond length changes and the optical spectra and kinetics have been measured and compared with those of ferrioxalate. Data presented here strongly suggest that both of these metal oxalato complexes operate under similar photoredox reaction mechanisms where the primary reaction involves the dissociation of a metal–oxygen bond. These results also indicate that excitation in the charge-transfer band is not a sufficient condition for the intramolecular electron transfer to be the dominant photochemistry reaction mechanism.

photoreduction | organometallic | ultrafast spectroscopy | time-resolved EXAFS | photodissociation

The photochemistry of transition metal trisoxalato complexes (1) has been studied extensively (2, 3), not only because of their wide application in areas such as chemical actinometry (4), radical polymerization reaction initiation (5), degradation of organic pollutants (6) and as solar energy media (7), but also because they have served as textbook models for electron transfer (ET) (8, 9) and stereochemistry (10). For a long period, transition metal trisoxalato complexes were thought to undergo exclusively intramolecular ET from the oxalate group to the metal, ligand to metal, immediately after irradiation inside the charge-transfer band. This hypothesis was based on continuous wave, flash photolysis (11–14) and nanosecond laser spectroscopic experimental results (15) in both aqueous and nonaqueous (16, 17) solutions. However, the proposed intramolecular ET process was thought to occur in the picosecond range, which could not be time-resolved with the methods that were then used. Owing to the lack of direct experimental support, such as transient absorption spectra or the observation of transient structural changes, intermolecular and intramolecular ET remained speculative. Is excitation in the charge-transfer band a sufficient condition for intramolecular electron transfer? What is the photochemical behavior difference between charge-transfer (CT) and ligand-field (LF) bands and why? The development of ultrafast spectroscopy, especially ultrafast x-ray spectroscopy (18–22), allow us to reevaluate the photochemical mechanism of transition metal complexes. Previously, we performed static extended x-ray absorption fine structure (EXAFS) spectroscopic experiments that revealed the structures of only the initial and final product in the photolysis of CBr_4 without any attempt, as clearly stated, to measure the structure of any intermediate product (23). This was in contrast to a report (24) that suggested that time-resolved EXAFS studies were performed and CBr_4 photolysis intermediates in solution were not observed. In fact, the final Br_3CCBr_3 product detected and measured (23) can only be formed by CBr_3 radical recombination. It is also to be noted that our time-resolved optical studies were aimed only at the cage intermediates (25) and not at the

study of the expected out-of-cage photochemical intermediates that were observed later by time-resolved x-ray diffraction (24). Recently, we reported on the photochemistry of ferrioxalate in water by means of ultrafast transient optical and EXAFS spectroscopy and density functional theory (DFT)/unrestricted Hartree–Fock (UHF) calculations (26–29). Previously, we reported that excitation of ferrioxalate with either 267/266 nm or 400/355 nm pulses results predominantly in Fe–O bond dissociation, concurrent with photoelectron detachment followed by electron solvation as a side reaction, rather than intramolecular ET (26, 27, 29). Direct intramolecular ET may take place with much lower efficiency. Solvated electrons were observed only by a two-photon process using 400-nm photons or a one-photon process using 267-nm excitation (26). This reaction path has also been observed for trisoxalato cobaltate (III) with similar photon energy selectivity (27). These results suggest that the strong absorption in the CT band is at least partially caused by the charge transfer from the trisoxalato metalate complex to the solvent. In the case of photodissociation, which we found to be the dominant reaction, we did not observe a significant difference in the kinetics and mechanism by exciting ferrioxalate either in the CT band, with 267-nm femtosecond (fs) pulses or the crossing point of the CT and LF band, with 400-nm fs pulses (26). These data indicate that excitation in the CT band does not necessarily yield intramolecular ET but rather is in competition with other reaction paths such as dissociation. Is this major reaction path restricted to ferrioxalate or does it also apply to other trisoxalato metal complexes such as trisoxalato cobaltate (III)? In this article, we present time-resolved kinetics and structure changes induced by 266/267-nm pulsed excitation, measured by means of femtosecond to microsecond transient optical spectroscopy and ultrafast picosecond EXAFS. In addition, we have performed DFT (B3LYP/6-31G), quantum chemical, and Hartree–Fock (H-F/6-31G) calculations that provide supporting information that has helped us to elucidate the mechanism of the photoredox reaction of trisoxalato cobaltate (III) in aqueous solution. The experimental results observed previously (26, 29) for ferrioxalate are also considered and compared with the trisoxalato cobaltate(III) to deduce a rather general mechanism of the photochemistry and ET of metal trisoxalato complexes.

Results

Ultrafast EXAFS Spectra. In the present study, 100 fs, 0.3 mJ, 267 nm, Ti:Sapphire 3rd harmonic pulses were used as the pump

Author contributions: J.C. and P.M.R. designed research; J.C. and H.Z. performed research; I.V.T. and X.D. contributed new reagents/analytic tools; J.C., H.Z., and P.M.R. analyzed data; and J.C., H.Z., and P.M.R. wrote the paper.

The authors declare no conflict of interest.

[§]To whom correspondence should be addressed. E-mail: pmrentze@uci.edu.

This article contains supporting information online at www.pnas.org/cgi/content/full/0806990105/DCSupplemental.

© 2008 by The National Academy of Sciences of the USA

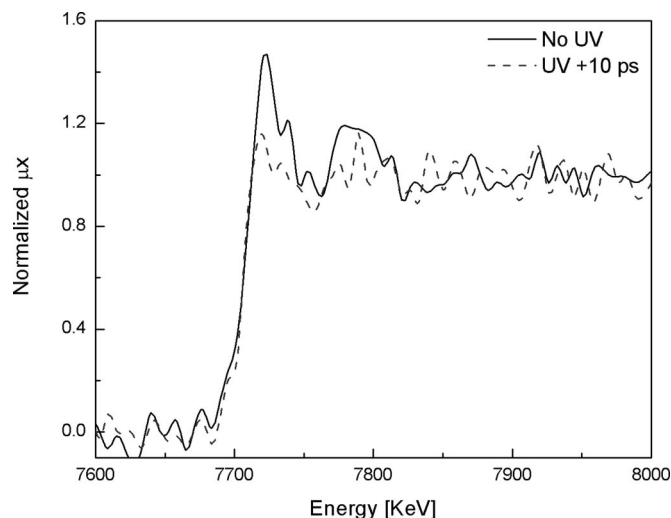


Fig. 1. EXAFS spectra of trisoxalato cobaltate (III)/water solution plotted as normalized $\mu\chi$ vs. energy: without UV (solid line) and 10 ps after 267-nm fs pulse excitation (dotted line).

pulses; 0.6 ps, 6.6–8.6 KeV x-ray pulses were used as the x-ray continuum probe pulses for time-solved transient structure EXAFS experiments. The continuum spectrum is shown in [supporting information \(SI\) Fig. S1](#). The k range is a bit limited owing to the $L_{\alpha 2}$ line; however, with long-time exposure experiments an acceptable fit has been possible. A broad-band plasma source based on femtosecond laser irradiation of a water jet in helium has been reported recently, which was free from characteristic emission lines but yielded a less intense continuum at this energy range (30). The energy resolution of the system was estimated to be 20 eV. This system has been used to measure the Fe–O bond lengths of the ferrioxalate redox reaction transients with 2-ps time resolution and 0.04 Å accuracy (26, 29). Time-resolved EXAFS spectra were obtained by focusing the x-ray pulses on the sample with a x-ray lens (31) and then collecting the absorption signal through an energy-dispersive spectrometer (32). The analysis of the EXAFS data were performed by using the standard automated data reduction program, ATHENA (33), and an *ab initio* multiple scattering calculations program for EXAFS and x-ray absorbance near-edge spectrum (XANES) spectra, FEFF 8.20 (34). The Co–O bond length was extracted from the EXAFS μx vs. energy spectra shown in Fig. 1, and presented in the form of $|\chi(R)|$ vs. R spectra (Fig. 2), which exhibits the bond distance between cobalt and oxygen of the first coordination shell. By using our time-resolved EXAFS experimental system, we determined that the Co(III)–O bond distances of the parent molecule in water at 10 ps before excitation and without excitation, were 1.89 Å and 1.90 Å, respectively, whereas the reported value obtained by steady-state EXAFS was 1.898 Å

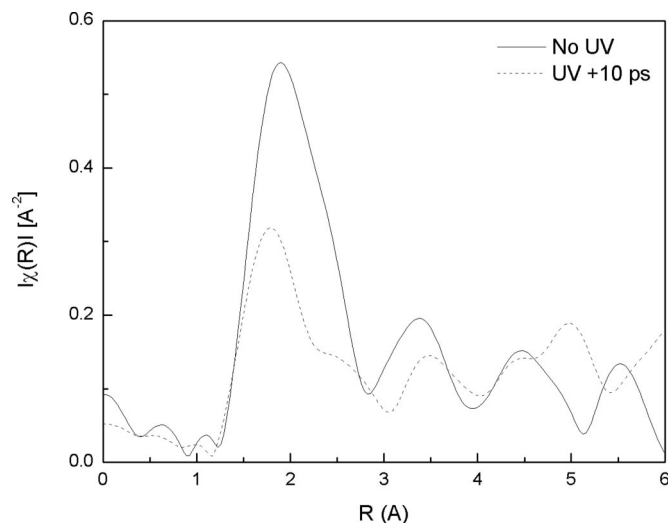


Fig. 2. R space EXAFS spectra of trisoxalato cobaltate (III)/water solution: without UV (solid line) and 10 ps after 267-nm UV radiation (dotted line).

(35). These experimentally measured bond distances are in good agreement with the 1.90-Å x-ray crystallographic literature value for the Co(III)–O bond distance of $[\text{Co(III)}(\text{C}_2\text{O}_4)_3]^{3-}$ (36). We also used DFT and UHF methods to calculate the structure of the ground-state molecule and both calculations yield a value of 1.92 Å for the Co–O bond distance. The changes of the Co–O bond length as a function of time during the first 142 ps are summarized in Table 1. Full geometry optimizations were performed for the ground state of each assigned structure by *ab initio* UHF and DFT calculations by using the Gaussian 03 program (37). The basis set 6-31G was used for all ground-state calculations. The Becke three-parameter hybrid functional with the Lee–Yang–Parr correlation corrections (B3LYP) was used in the DFT calculations. Some theoretical results for ferrioxalate have also been reported (26, 28). The very good agreement between theoretical calculations and the experimental data made it possible to propose a mechanism that is consistent with these data and is also supported by additional optical and radical scavenging experimental results. Our results of the ground-state structure calculations of the original molecule and transients and their assignment are summarized in Table 1. For the ground state of the parent Co(III) complexes, we used the $S = 0$ low spin state, which has been verified experimentally (38), and for the Co(II) complexes, the $S = 3/2$ high spin state was used, which agrees with the magnetic susceptibility measurements of $\text{K}_2\text{Co(II)}(\text{C}_2\text{O}_4)_2$ (39).

Optical Transient Absorption Spectra. The laser systems that were used to determine the femtosecond, picosecond, and nanosecond optical transients of trisoxalato cobaltate (III) have been

Table 1. Metal–oxygen bond length at various delay times before and after 267-nm fs pulse excitation obtained by time-resolved EXAFS and DFT/UHF quantum chemistry calculations

Assignment	Ligand	M = Fe			M = Co		
		Delay time, ps	Exp. R, Å	DFT (UHF) Cal. R, Å	Delay time, ps	Exp. R, Å	DFT (UHF) Cal. R, Å
[M(III)(C ₂ O ₄) ₃] ^{3−}	C ₂ O ₄	−20	1.99	2.01 (2.04)	−10	1.89	1.92 (1.92)
[M(C ₂ O ₄) ₃] ^{3−*}	C ₂ O ₄	0–2	2.21	N/A	0	1.98	N/A
[C ₂ O ₃ O] − M(III)(C ₂ O ₄) ₂] ^{3−}	C ₂ O ₃ O	4	1.92	1.87 (1.87)	2	1.93	1.83 (1.83)
	C ₂ O ₄			2.02 (2.01)			1.86–1.93
							(1.86–1.92)
[M(III)(C ₂ O ₄) ₂] [−] Tetrahedral-like	C ₂ O ₄	5–140	1.89–1.93	1.90 (1.90)	4–142	1.78–1.81	1.81–1.84 (1.81)

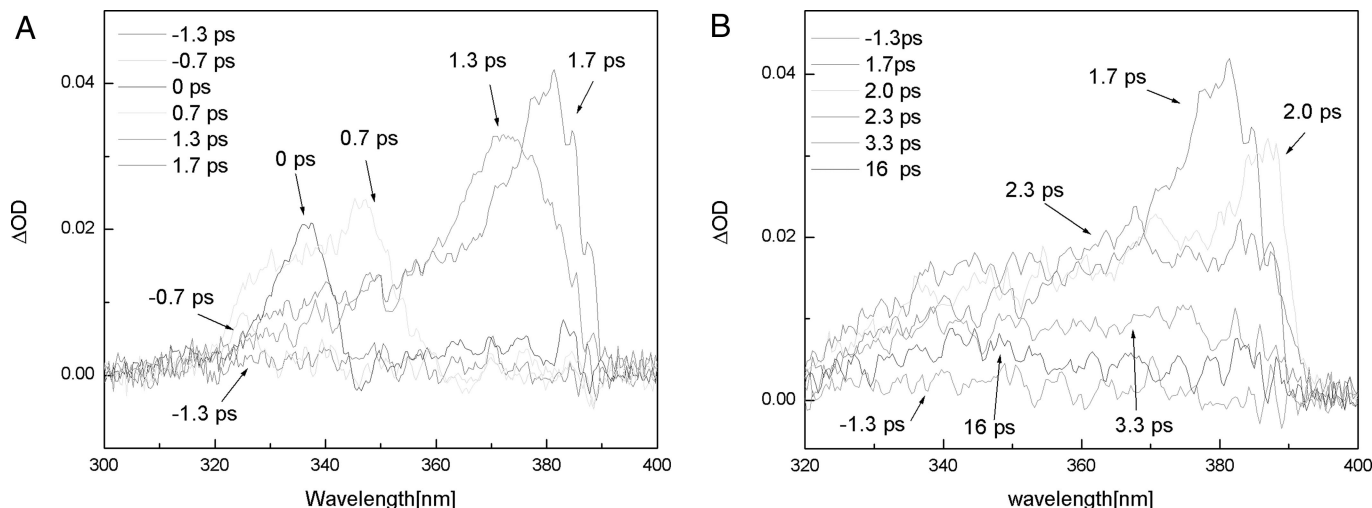


Fig. 3. Femtosecond time-resolved transient absorption spectra of 1.0×10^{-3} M trisoxalato cobaltate (III) in water by using 267-nm excitation: from -1.3 to 1.7 ps (A); from -1.3 to 16 ps (B).

described (26, 29). For the present studies, the pump pulses consisted of 100 fs, 0.3 mJ, 267-nm pulses (3rd harmonic of Ti:Sapphire laser) or 35 ps, 7 ns, 1 mJ 266-nm pulses (4th harmonic of the Nd:YAG laser). The probe continuum was generated by focusing the 800-nm fundamental or 400-nm 2nd harmonic of a Ti:Sapphire laser in a 5-mm H_2O cell. The trisoxalato cobaltate (III) concentrations used varied from 0.28 to 2.3 mM. Our transient optical data show that after excitation with 267-nm fs pulses, two intermediate absorption bands were formed at 340–390 nm (see Fig. 3) and at 400–800 nm (shown in *SI Text* and Figs. S2–S4). The formation and decay kinetics of the 340- to 390-nm transient band is depicted in Fig. 4 and is summarized in Table 2. Immediately after excitation with 20-ns 266-nm pulses, two absorption bands were observed (Fig. 5), one located between 340 nm and 400 nm and the other in the 400- to 800-nm range. The 400–800 nm is a solvated electron band that decays exponentially with a lifetime of 72 ns and disappears 500 ns after excitation, whereas the 340- to 400-nm band with a maximum wavelength at 360 nm is still evident. Nanosecond and microsecond transient spectra at 5.8×10^{-4} M show that this

transient absorption band has a two-component, diffusion-controlled decay, a short component with a 55-ns decay lifetime and a long component with a 55- μs decay lifetime (see Fig. S5). At 500 ns after excitation, two bands with negative ΔOD were also observed at 420 nm and 600 nm, respectively. These bands have exactly the same wavelength range and maxima as the absorption band of the nonexcited trisoxalato cobaltate (III). They were not observed earlier than 500 ns after excitation, because they were masked by the intense 400- to 800-nm solvated electron absorption band and became evident after 500 ns when the solvated electron band decayed. Therefore, we attribute them to the bleaching of the ground state of trisoxalato cobaltate (III).

Discussion

Photodissociation. Ultrafast optical studies. Femtosecond transient absorption spectra (Fig. 3A) show that after excitation with a 267-nm fs pulse, an absorption band was formed in the 320- to 390-nm range that undergoes the fast decay shown in Fig. 3B. The continuous band shift of the maxima from 0 to 1.3 ps depicted in Fig. 3A is due to group velocity dispersion of the probe 320- to 390-nm continuum, and the spectra corrected for dispersion are shown in Fig. S4. The kinetics of this transient absorption band at 380 nm plotted in Fig. 4 in the form of ΔOD vs. t show an OD increase, at 380 nm, from 0 to 0.04 OD within 1 ps, followed by a rapid decrease from 0.04 OD to 0.007 OD within 1.6 ps, corresponding to a decay lifetime of 0.8 ps. The decay may be attributed to the vibronic relaxation internal conversion within the excited state, which is typical of large molecules in the condensed phase. After 3.3 ps, the 380-nm intensity remains constant for at least 20 ps.

Ultrafast EXAFS studies. Based on our ultrafast EXAFS data obtained for trisoxalato cobaltate (III), we determined that the Co–O bond length has a value of 1.90 Å, in the original nonirradiated form, then increases to 1.98 Å immediately after excitation, followed by a decrease to 1.93 Å after 2 ps, and then a further decrease to 1.78 Å after 4 ps. At times longer than 4 ps the Co–O bond length was measured to have a value of 1.81 Å and remained as such for the 142-ps time span capability of our femtosecond system. The trend of these observed bond length changes for trisoxalato cobaltate (III) is very similar to that observed for ferrioxalate under the same experimental conditions, namely 1.99 Å for the parent molecule, 2.21 Å at +2 ps, 1.92 Å after 4 ps, and 1.89–1.93 Å for 5–140 ps (see Table 1 for a list of both trisoxalato metalates). Based on the similarity of the

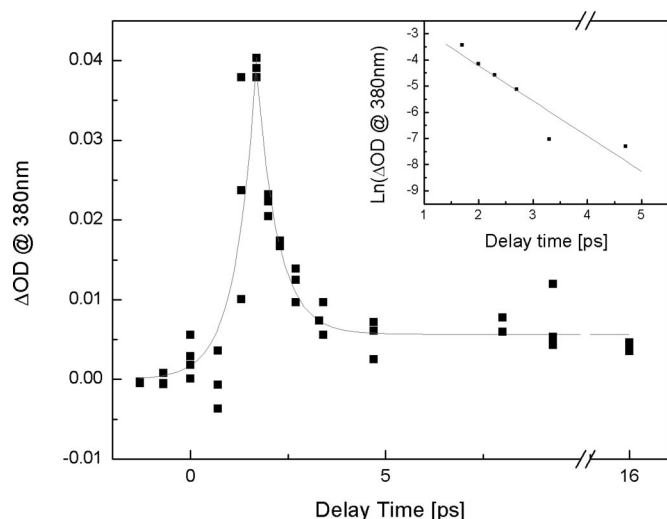
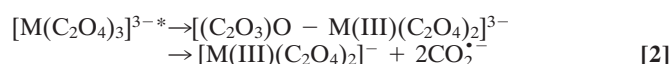


Fig. 4. Femtosecond kinetics of 1.0×10^{-3} M trisoxalato cobaltate (III) in water at 380 nm after 267-nm excitation. (Inset) Semilog plot of the transient optical density at 380 nm.

Table 2. Kinetics of optical transient spectra of trisoxalato cobaltate (III) aqueous solution using 267/266 nm excitations

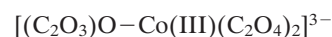
Time	Spectra (nm)	$\tau_{\text{formation}}$	τ_{decay}	Assignment
ps	320–390	< 1 ps at 380 nm	0.8 ps at 380 nm	CTTS state of $[\text{Co(III)}(\text{C}_2\text{O}_4)_3]^{3-}$
	400–800	< 1.3 ps at 720 nm	25 ps at 720 nm	e_{aq}^-
ns	340–400	Within 20-ns pulse width	55 ns at 363 nm	$[\text{Co(III)}(\text{C}_2\text{O}_4)_2]^-$
	400–800		72 ns at 720 nm	e_{aq}
μs	340–400		55 μs at 363 nm	$[\text{Co(II)}(\text{C}_2\text{O}_4)_3]^{4-}$

structural changes of trisoxalato cobaltate (III) and ferrioxalate, we propose the following reaction mechanism for both molecules.



We attribute the structural changes observed for trisoxalato cobaltate (III) to the following transient species: (i) -10 ps: original ground-state nonexcited trisoxalato cobaltate (III); the Co–O bond length is 1.89 Å. (ii) 0 ps: excited state. The Co–O bond length measured at 0 ps after 267 -nm excitation by our ultrafast EXAFS system was found to be 1.98 Å. This 1.98 -Å Co–O bond length is attributed to an excited state of trisoxalato cobaltate, $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-*}$, which has been elongated by 0.08 Å compared with the ground-state molecule. A similar bond length increase by ≈ 0.09 Å has also been observed lately in the Fe–N bond after excitation (40). Although theoretical calculations for the excited-state structure are not available, we analyzed the electronic properties of the interacting frontier molecular orbitals, highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO), at the optimized ground-state geometry. The sum of the squares of the MO coefficients of the total atomic contributions from Co and three oxalate groups are 0.34 and 0.61 , respectively, in HOMO. Those are changed to 0.70 for Co and 0.41 for oxalate, respectively, in LUMO. Those calculation results show (i) the extent of mixing of Co and oxalate orbitals in HOMO and

LUMO; *(ii)* the cobalt character increasing from 0.34 in HOMO to 0.70 in LUMO, which represents a partial charge transfer from oxalate to cobalt; *(iii)* a 2-ps, five-coordinated Co(III) oxalate complex.—The Co–O bond length obtained by using 267-nm excitation became 1.93 Å after 2 ps, and is assigned to



five-coordinate complex. We proposed that this intermediate is formed after breaking one Co–O bond. The Co–O bond distances calculated by DFT for this five-coordinate complex are 1.84 Å for one bond and 1.97–2.02 Å for the remaining four Co–O bonds.—and (iv) a 4–142 ps, four-coordinated Co(III) oxalate complex. The Co–O bond distances listed in Table 1 were determined to be 1.78 Å at 4 ps after 267-nm excitation and 1.81 Å after 10 ps and remained at 1.81 Å for the 142-ps limit of our EXAFS experiments. The time and spatial resolution of our ultrafast x-ray system is 2 ps and 0.04 Å, respectively (26). The 1.78–1.81 Å Co–O bond length is assigned to the [Co(III)(C₂O₄)₂][−] four-coordinated dissociation product (28). Theoretical calculations show that the Co–O bond length of [Co(III)(C₂O₄)₂][−] is 1.80 Å, which agrees very well with our 1.78–1.81 Å experimental value. This assignment assumes that the dissociation product [Co(III)(C₂O₄)₂][−] remains in the + 3 oxidation state, and suggests that intramolecular ET from oxalate to cobalt is not the dominant reaction during this time period, although we do not exclude its involvement.

We also considered the mechanism of intramolecular ET by calculating the structure of $[\text{Co(II)(C}_2\text{O}_4)_2]^{2-}$ ($S = 3/2$). These DFT calculations show that the $[\text{Co(II)(C}_2\text{O}_4)_2]^{2-}$ ion has a tetrahedral-like configuration and a Co–O bond length of 1.98 Å. This bond length looks similar to the bond length that was observed at 0 ps after excitation. If we assumed that the species we observed, just after excitation, was the Co (II) complex, which might have a similar Co(II)–O bond distance as $[\text{Co(II)(C}_2\text{O}_4)_2]^{2-}$, then it becomes difficult to understand the process that proceeds from Co(III) to Co(II) and then returns back to Co(III) complex. The change from Co(II) to Co(III) complex is possible in the nanosecond range when the oxygen dissolved in the water solution initiates a diffusion-controlled oxidation reaction, but is not a likely reaction during the 2- to 4-ps EXAFS time range that we investigated. We note that the elongation of the Co–O bond length from 1.90 Å in the ground state to the excited state by 0.08 Å suggests that the excited state may indeed have Co(II) character to a certain extent. This is understandable by an intramolecular charge-transfer mechanism from oxygen to cobalt. However, this partial charge transfer might lead to two reaction paths: (path 1) intramolecular electron transfer and (path 2) breaking up a Co–O bond facilitated by the bond elongation. Which of these two mechanisms is dominant depends on the reaction rate of each path. The Co–O bond lengths obtained in the 4- to 142-ps range indicate that path 2, dissociation of Co–O bond, is dominant. However, path 1 is not ruled out. The structural changes that occur at times longer than 142 ps after excitation could not be measured by our present subpicosecond EXAFS system. However, the transient

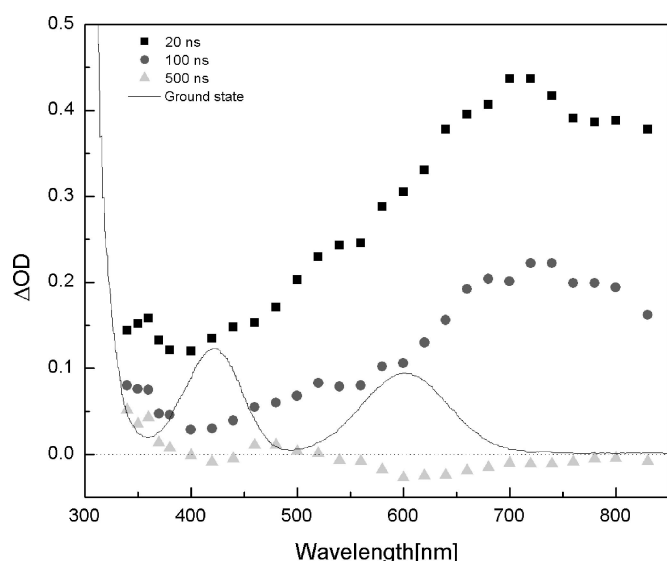
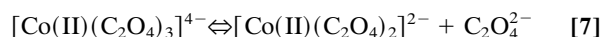
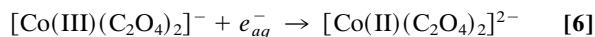
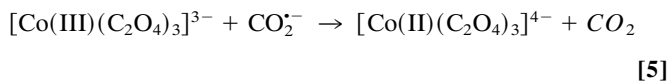
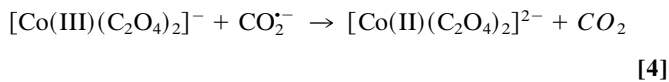
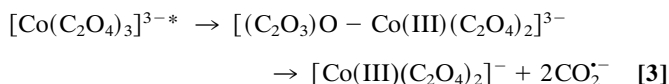


Fig. 5. Transient absorption spectra of trisoxalato cobaltate(III) in water by using 266-nm excitation at different delay times ($c = 5.8 \times 10^{-4}$ M): 20 ns (squares), 100 ns (circles), 500 ns (triangles), and ground-state absorption spectrum (line).

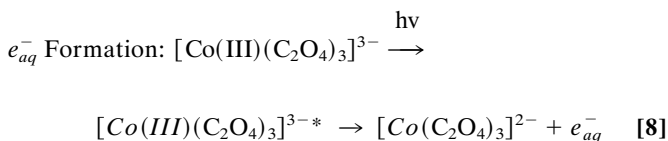
optical spectroscopic data that we have presented were used to determine the reaction mechanism at times >142 ps.

Intermolecular electron transfer. The photodissociation path is the result of direct dissociation of trisoxalato cobaltate (III) by a 267/266 nm photon. One oxalate ligand is dissociated, resulting in the generation of carbon dioxide radical anions $\text{CO}_2^{\cdot-}$ and $[\text{Co(III)}(\text{C}_2\text{O}_4)_2]^-$. The newly formed $\text{CO}_2^{\cdot-}$ may reduce the dissociated product $[\text{Co(III)}(\text{C}_2\text{O}_4)_2]^-$, reaction 5, or parent molecule $[\text{Co(III)}(\text{C}_2\text{O}_4)_3]^{3-}$, reaction 6, by intermolecular electron transfer. The 363-nm ns transient band shown in Fig. 5 is assigned to $[\text{Co(III)}(\text{C}_2\text{O}_4)_3]^{3-}$. The decay lifetime of this transient is found to be in the nanosecond range and is concentration-dependent, which is mostly due to the diffusion-controlled reaction between $[\text{Co(III)}(\text{C}_2\text{O}_4)_2]^-$ and $\text{CO}_2^{\cdot-}$, reaction 4, and to a lesser extent to the reaction between $[\text{Co(III)}(\text{C}_2\text{O}_4)_2]^-$ and solvated electron depicted in reaction 6. After $\text{CO}_2^{\cdot-}$ reacts with $[\text{Co(III)}(\text{C}_2\text{O}_4)_3]^{3-}$, the reduced product $[\text{Co(II)}(\text{C}_2\text{O}_4)_3]^{4-}$ releases one oxalate and is in equilibrium with the final product $[\text{Co(II)}(\text{C}_2\text{O}_4)_2]^{2-}$, whose reaction rate was determined to be in the microsecond range and also concentration-dependent. The equilibrium, reaction 7, will obviously shift to the left when the concentration of oxalate increases and will result in a slower decay of $\text{Co(II)}(\text{C}_2\text{O}_4)_3^{4-}$ (see Fig. S5). Radical scavenger experiments were also performed to identify the presence and involvement of $\text{CO}_2^{\cdot-}$ radical anion in the electron transfer process. Yet, a very weak scavenger effect was found (see SI Text).

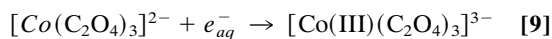


Photoelectron Detachment. A 400- to 800-nm transient was generated by one-photon, 267-nm, excitation of trisoxalato cobaltate (III) and is assigned to solvated electrons (27) for the following reasons. (i) It has the shape, width, spectral range, and absorption maximum (41, 42). (ii) Our experimental 1.3-ps formation lifetime, reaction 8, is very similar to refs. 41 and 42. (iii) The biexponential decay lifetime components are composed of: (a) The ≈ 25 -ps lifetime of newly formed solvated electron that recombines with the dissociated product $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ in the solvation cage to form the original species, reaction 9 (27). (b) The solvated electrons that escape the cage may react with the parent $[\text{Co(III)}(\text{C}_2\text{O}_4)_3]^{3-}$ molecule by a nanosecond concentration-dependent reaction 10 (see Table 2). (iv) The decay lifetime depends on the concentration of the nitrate electron scavenger. The experimentally measured bimolecular quenching constant of $1.4\text{--}1.5 \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$ agrees well with the literature values (43). Based on our data, the following photochemical mechanism is proposed: 267/266 nm excitation of trisoxalato cobaltate (III) in addition to dissociation generates

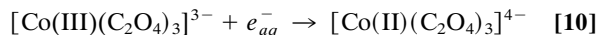
solvated electrons. The reaction path for the electrons may include cage recombination and/or reaction with parent molecules $[\text{Co(III)}(\text{C}_2\text{O}_4)_3]^{3-}$:



Cage recombination:



Out cage reaction:



The 720-nm band is assigned to the solvated electron transient and the strong absorption band of trisoxalato cobaltate (III) between 200 nm and 350 nm shown in Fig. 4 is assigned to be a charge transfer to solvent (CTTS) absorption band. The electron/ $[\text{Co}(\text{C}_2\text{O}_4)_3]^{2-}$ reaction time, 25 ps, is at least one order magnitude longer than the 0.8-ps, 380-nm band decay lifetime, therefore, this reaction cannot be considered to be responsible for the 380-nm fast-decay lifetime. However, the 267-nm photon can excite the trisoxalato cobaltate (III) molecules to the CTTS states, which decay with a lifetime of several hundred femtoseconds (44). This decay lifetime agrees well with our 0.8-ps experimental data for the decay of the 380-nm band. Therefore, we attribute the 380-nm transient with a 0.8-ps decay lifetime to CTTS states of the trisoxalato cobaltate (III) molecule. The quantum yield of the solvated electron formation is estimated by comparing the intensity of the transient absorption bands of $\text{Co(III)}(\text{ox})$ complex and ferrocyanide solutions at 680 nm under the same experimental conditions. By using the quantum yield of solvated electron generated in ferrocyanide, which is reported to be ≈ 1 (45), we determined the quantum yield for the photoelectron detachment from trisoxalato cobaltate (III) to be ≈ 0.10 . The quantum yield of $\text{Co(II)}(\text{ox})$ formation has been measured to be 0.7 (46); therefore, the photoelectron detachment is a low quantum yield side reaction. Similar results have been observed for ferrioxalate (29).

In summary, based on the time resolved EXAFS Co–O bond distances, the transient optical data and the DFT/UHF theoretical calculations we propose that the dominant photoredox reaction of trisoxalato cobaltate (III) initiated by 267/266 nm excitation in the charge-transfer band is a fast dissociation process, rather than intramolecular electron transfer, with photoelectron detachment and subsequent solvated electron processes as side reactions.

Materials and Methods

$(\text{NH}_4)_3\text{Co(III)}(\text{ox})_3 \cdot 3.5\text{H}_2\text{O}$, where $\text{ox} = \text{C}_2\text{O}_4^{2-}$, was prepared and purified according to ref. 47. The absorption extinction coefficient of trisoxalato cobaltate (III) at 267 nm was $1.6 \times 10^4 \text{ cm}^{-1} \text{ M}^{-1}$. The concentration used for time-resolved EXAFS experiments was 1.0 M, which corresponds to $\mu_{\text{X}} \approx 1.0$ for 7.7 KeV radiation. Precise spatial and temporal overlap of the 267-nm optical pump beam and the 6.6–8.6 KeV x-ray probe beam was achieved by using the procedure described in refs. 26 and 29.

ACKNOWLEDGMENTS. This work was supported in part by National Science Foundation Grant CHE-0079752 and W. M. Keck Foundation.

- Krishnamurthy KV, Harris GM (1961) The chemistry of the metal oxalato complexes. *Chem Rev* 61:213–246.
- Porter GB, Doering JGW, Karanka S (1962) Photolysis of transition metal oxalato complex ions. *J Am Chem Soc* 84:4027–4029.
- Stasicka Z, Marchaj A (1977) Flash-photolysis of coordination-compounds. *Coord Chem Rev* 23:131–181.

- Hatchard CG, Parker CA (1956) A new sensitive chemical actinometer. 2. Potassium ferrioxalate as a standard chemical actinometer. *Proc R Soc Ser A* 235:518–536.
- Baumann H, Strehmel B, Timpe HJ (1984) Light-induced polymer and polymerization reactions. 9. Initiation of photopolymerization by the initiator systems trisoxalatoferate arene onium salts. *Polym Photochem* 4:223–240.

6. SafarzadehAmiri A, Bolton JR, Cater SR (1997) Ferrioxalate-mediated photodegradation of organic pollutants in contaminated water. *Water Res* 31:787–798.
7. Heidt LJ, McMillan AF (1954) Influence of perchloric acid and cerous perchlorate upon the photochemical oxidation of cerous to ceric perchlorate in dilute aqueous perchloric acid. *J Am Chem Soc* 76:2135–2139.
8. Balzani V, Carassiti V (1970) *Photochemistry of Coordination Compounds* (Academic, New York), pp 221–224.
9. Endicott JF (1975) Charge-transfer photochemistry. *Concepts of Inorganic Photochemistry*, eds Adamson AW, Fleischauer PD (Wiley, New York), pp 81–142.
10. Inoue Y, Ramamurthy V (2004) *Chiral Photochemistry* (Marcel Dekker, New York).
11. Parker CA, Hatchard CG (1959) Photodecomposition of complex oxalates—Some preliminary experiments by flash photolysis. *J Phys Chem* 63:22–26.
12. Cooper GD, DeGraff BA (1971) On the photochemistry of the ferrioxalate system. *J Phys Chem* 75:2897–2902.
13. Cordeman L, Doliesla J, Hendrix J, Dejaeger S (1974) Flash-photolysis of potassium tris(oxalato)cobaltate(III). *J Phys Chem* 78:1361–1367.
14. Rowan NS, Hoffman MZ, Milburn RM (1974) Intermediates in photochemistry of tris(oxalato)cobaltate(III) ion in aqueous-solution—Free and coordinated radicals. *J Photochem Photobiol A* 99:145–153.
15. Nadtochenko V, Kiwi J (1996) Dynamics of light-induced excited state quenching of ferrioxalate complexes by peroxides. Fast kinetic events and interaction with toxic pollutants. *J Photochem Photobiol A* 99:145–153.
16. Ionescu SG, Oncescu T (1983) Photolysis of $K_3[Fe(C_2O_4)_3]$ in dimethylformamide. *J Photochem* 23:45–50.
17. Joosten M, D'Olieslager J (1998) Alternative reaction path for the photochemical reaction of tris(oxalato)cobaltate(III) in various aprotic solvents in the presence of oxygen. *Inorg Chim Acta* 271:29–35.
18. Raksi F, et al. (1996) Ultrafast x-ray absorption probing of a chemical reaction. *J Chem Phys* 104:6066–6069.
19. Bressler C, Chergui M (2004) Ultrafast x-ray absorption spectroscopy. *Chem Rev* 104:1781–1812.
20. Pfeifer T, Spielmann C, Gerber G (2006) Femtosecond x-ray science. *Rep Prog Phys* 69:443–505.
21. Gaffney KJ, Chapman HN (2007) Imaging atomic structure and dynamics with ultrafast X-ray scattering. *Science* 316:1444–1448.
22. Chen LX, et al. (2007) Tracking electrons and atoms in a photoexcited metalloporphyrin by X-ray transient absorption spectroscopy. *J Am Chem Soc* 129:9616–9618.
23. Oulianov DA, Tomov IV, Dvornikov AS, Rentzepis PM (2002) Structures of bromoalkanes' photodissociation in solution by means of ultrafast extended x-ray absorption fine-structure spectroscopy. *Proc Natl Acad Sci USA* 99:12556–12561.
24. Kong Q, et al. (2007) Photochemical reaction pathways of carbon tetrabromide in solution probed by picosecond x-ray diffraction. *J Am Chem Soc* 129:13584–13591.
25. Zhang H, Dvornikov AS, Rentzepis PM (2005) Photolysis of CBr_4 and its transient solvent stabilized ($CBr_3^+//Br^-$)(Solv) ion pair. *J Phys Chem A* 109:5984–5988.
26. Chen J, et al. (2007) Transient structures and kinetics of the ferrioxalate redox reaction studied by time resolved EXAFS, optical spectroscopy and DFT. *J Phys Chem A* 111:9326–9335.
27. Zhang H, et al. (2007) Photoelectron detachment and solvated electron dynamics of the cobalt(III) and iron(III) oxalato complexes. *J Phys Chem A* 111:11584–11588.
28. Chen J, et al. (2007) Electron transfer and dissociation mechanism of ferrioxalate: A time resolved optical and EXAFS study. *Chem Phys Lett* 437:50–55.
29. Chen J, Zhang H, Tomov IV, Rentzepis PM (2008) Electron transfer and photochemistry of ferrioxalate induced by excitation in the charge transfer band. *Inorg Chem* 47:2024–2032.
30. Fullagar W, et al. (2007) A broadband laser plasma x-ray source for application in ultrafast chemical structure dynamics. *Rev Sci Instrum* 78:115105.
31. Sun TX, Liu ZG, Ding XL (2007) Characterization of a polycapillary focusing X-ray lens for application in spatially resolved EXAFS experiments. *Chem Phys Lett* 439:412–414.
32. Oulianov DA, Tomov IV, Lin SH, Rentzepis PM (2001) Time-resolved extended x-ray absorption fine structure (EXAFS) studies by means of an energy dispersive spectrometer. *J Chin Chem Soc* 48:127–132.
33. Ravel B, Newville M (2005) ATHENA, ARTEMIS, HEPHAESTUS: Data analysis for X-ray absorption spectroscopy using IFFFIT *J Synchro Rad* 12:537–541.
34. Ankudinov AL, et al. (2002) Parallel calculation of electron multiple scattering using Lanczos algorithms. *Phys Rev B* 65:104107.
35. Sakane H, et al. (1994) Solvent effects on cobalt(III) and chromium(III) oxalato complexes by x-ray-absorption fine-structure. *Polyhedron* 13:1625–1630.
36. Okazaki H, Kushi Y, Yoneda H (1985) Spontaneous resolution and polymorphism of potassium tris(oxalato)cobaltate(III)—X-ray crystallographic and thermal studies. *J Am Chem Soc* 107:4183–4189.
37. Frisch MJ, et al. (2004) Gaussian 03 (Gaussian, Inc., Wallingford, CT), Revision C.02.
38. Ballhausen CJ, Asmussen RW (1957) The diamagnetic susceptibility and the high-frequency term in Co(III)-complex and Rh(III)-complexes. *Acta Chem Scand* 11:479–483.
39. Field FH, Vosburgh WC (1949) Complex ions. 9. Magnetic susceptibilities of nickel and cobalt complex ions from 0-degrees to 80-degrees. *J Am Chem Soc* 71:2398–2401.
40. Gawelda W, et al. (2007) Structural determination of a short-lived excited iron(II) complex by picosecond x-ray absorption spectroscopy. *Phys Rev Lett* 98:057401.
41. Baxendale JH, et al. (1964) Pulse radiolysis. *Nature* 201:468–470.
42. Pommeret S, et al. (1998) Ultrafast events in the electron photodetachment from the hexacyanoferrate(II) complex in solution. *Chem Phys Lett* 288:833–840.
43. Jonah CD, Miller JR, Hart EJ, Matheson MS (1975) Picosecond pulse-radiolysis. 1. Time or concentration dependent rate constants. *J Phys Chem* 79:2705–2711.
44. Long FH, Lu H, Shi XL, Eisenthal KB (1990) Femtosecond studies of electron photodetachment from an iodide-ion in solution—The trapped electron. *Chem Phys Lett* 169:165–171.
45. Wiesenfeld JM, Ippen EP (1980) Dynamics of electron solvation in liquid water. *Chem Phys Lett* 73:47–50.
46. Copestake TB, Uri N (1955) The photochemistry of complex ions—Photochemical and thermal decomposition of the trioxalatocobaltate-III complex. *Proc R Soc Lond Math Phys Sci* 228:252–263.
47. Willard HH, Furman NH, Bricker CE (1956) in *Elements of Quantitative Analysis* (Van Nostrand, New York), pp 250–251.